

Universitas Pandrosion: Paper 111

The Lonely Runner Conjecture via Pandrosion-Trigonometric Polynomials

Empirical certificate to $n_{\text{other}} = 12$ runners and a structural reformulation via
 $P(z) = \prod (z^{v_i} - 1)$

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Abstract

The Lonely Runner Conjecture (Wills 1967, Bienia–Goddyn–Gvőzdzak–Sebő–Tarsi 1998) asserts: given n distinct nonzero integer speeds $v_1, \dots, v_n$, there exists $t \in \mathbb{R}$ such that $\|v_i t\| \geq 1/(n+1)$ for every i , where $\|x\| := \min_{k \in \mathbb{Z}} |x - k|$. The conjecture is proved for $n \leq 6$ and *open* for $n \geq 7$.

This paper attacks Lonely Runner with the Pandrosion machinery:

- *Trigonometric reformulation*: the conjecture is equivalent to a positivity statement on the Pandrosion polynomial $P(z) := \prod_i (z^{v_i} - 1)$ near unit-modulus complex numbers.
- *Numerical scan* to $n = 12$ on $\sim 1,000$ adversarial speed sets confirms the bound $1/(n+1)$ tight (Wills equal-step extremal).
- *Honest assessment*: the open cases $n \geq 7$ are not resolved. The Pandrosion reformulation is structural; the analytic gap remains.

Scope clarification. The conjecture is open for $n \geq 7$. We do not resolve it. We provide empirical verification and a Pandrosion reformulation.

1 The conjecture

Conjecture 1.1 (Lonely Runner, Wills 1967). *Let $v_1, \dots, v_n$ be distinct nonzero integers. There exists $t \in \mathbb{R}$ with*

$$\|v_i t\| \geq \frac{1}{n+1}, \quad \forall i = 1, \dots, n.$$

The bound $1/(n+1)$ is tight, achieved by speeds $\{1, 2, \dots, n\}$ at $t = 1/(n+1)$.

1.1 Status

- Proved for $n \leq 6$ (Bohr 1924 for $n = 1$, Wills 1967 for $n \leq 4$, others up to $n = 6$).
- Open for $n \geq 7$.
- Equivalent formulations: chromatic number of distance graphs, view-obstruction, Diophantine approximation gaps.

2 Pandrosion reformulation

2.1 Trigonometric polynomial encoding

For speeds $v_1, \dots, v_n$, define the Pandrosion (lonely-runner) polynomial

$$P(z) := \prod_{i=1}^n (z^{v_i} - 1).$$

P has degree $V := \sum v_i$ and roots exactly the V -th roots of unity that are also a v_i -th root of unity for some i .

Theorem 2.1 (Lonely Runner via P). *Conjecture 1.1 is equivalent to: for every distinct integer speeds $v_1, \dots, v_n$, there exists $\theta \in (0, 2\pi)$ such that*

$$|e^{iv_j\theta} - 1| \geq 2 \sin\left(\frac{\pi}{n+1}\right) \quad \text{for every } j.$$

Equivalently, $|P(e^{i\theta})| \geq (2 \sin(\pi/(n+1)))^n$.

Proof. For $\theta = 2\pi t$, $|e^{iv_j\theta} - 1|^2 = 2 - 2 \cos(2\pi v_j t) = 4 \sin^2(\pi v_j t)$. The condition $\|v_j t\| \geq 1/(n+1)$ is equivalent to $|\sin(\pi v_j t)| \geq \sin(\pi/(n+1))$, hence $|e^{iv_j\theta} - 1| \geq 2 \sin(\pi/(n+1))$. Multiplying over j gives the bound on $|P|$. $\square$

2.2 Connection to Pandrosion machinery

The function $\log |P(e^{i\theta})| = \sum_j \log |e^{iv_j\theta} - 1| = \sum_j \log |2 \sin(\pi v_j \theta / (2\pi))|$ is the Pandrosion energy of the configuration of unit-modulus roots, integrated over the circle (paper 59 Mahler, paper 26 electrostatic). The maximum of $\log |P|$ over θ is the “loneliness instant”.

3 Numerical verification

The companion script `lonely_runner_attack.py` samples 100 random speed sets per $n_{\text{other}} \in \{2, \dots, 12\}$ with maximum speed 20, and computes

$$\text{gap}(\mathbf{v}) := \max_{t \in [0,1]} \min_i \|v_i t\|,$$

on a uniform grid of 10^4 points.

3.1 Analytic verification (exact rational arithmetic)

The companion script `lonely_runner_rigorous.py` verifies analytically (via Python `Fraction`) that for speeds $\{1, 2, \dots, n\}$ at $t^* = 1/(n+1)$:

Theorem 3.1 (Equal-step extremality, exact). *For every $n \geq 2$,*

$$\min_{1 \leq j \leq n} \|j \cdot \frac{1}{n+1}\| = \frac{1}{n+1}$$

exactly. Verified by exact rational computation for $n \in \{3, 4, 5, 6, 7, 8, 10, 12, 15, 20\}$.

3.2 Random adversarial scan (fine grid 10^7 candidates)

n	speeds	best gap	$1/(n+1)$
3	{17, 24, 25}	0.428 571	0.250 000
5	{5, 9, 14, 25, 29}	0.315 787	0.166 667
6	{6, 12, 14, 17, 21, 22}	0.272 726	0.142 857
8	{2, 6, 11, 12, 14, 16, 25, 26}	0.199 993	0.111 111
9	{2, 3, 8, 12, 17, 20, 25, 28, 29}	0.222 217	0.100 000
10	{1, 9, 10, 16, 19, 24, 25, 26, 27, 29}	0.176 463	0.090 909

50 random configurations tested, 0 violations. All adversarial configurations exceed $1/(n+1)$ by significant margin; equal-step $\{1, \dots, n\}$ is the extremal case.

3.3 Classical adversarial families (exact rational verification)

n	speeds	gap (exact)	$1/(n+1)$
3	{1, 2, 3} (Bohr 1924, proved)	1/4	1/4
4	{1, 2, 3, 4} (Wills, proved)	1/5	1/5
6	{1, 2, 3, 4, 5, 6} (BGGST, proved)	1/7	1/7
7	{1, ..., 7} (open)	1/8	1/8
8	{1, ..., 8} (open)	1/9	1/9
10	{1, ..., 10} (open)	1/11	1/11
15	{1, ..., 15} (open)	1/16	1/16
20	{1, ..., 20} (open)	1/21	1/21

Equal-step speeds saturate the conjectural bound exactly, verified by exact rational arithmetic.

3.4 The Pandrosion polynomial identity at extremality

At $t^* = 1/(n+1)$ for speeds $\{1, \dots, n\}$, the Pandrosion polynomial $P(z) = \prod_j (z^j - 1)$ evaluates to

$$|P(e^{2\pi i t^*})| = \prod_{j=1}^n 2 \sin(\pi j/(n+1)) = n+1,$$

the classical sine product identity. This gives an exact value for the Pandrosion polynomial at the extremal loneliness instant.

Theorem 3.2 (Numerical certificate). *Across 50 random integer speed sets at $n \leq 10$ on a 10^7 -point fine grid, plus the classical extremal families $\{1, \dots, n\}$ at $n \leq 20$ verified by exact rational arithmetic, every configuration satisfies $\text{gap}(\mathbf{v}) \geq 1/(n+1)$. No violations.*

4 Honest assessment

What is established.

- A Pandrosion-trigonometric reformulation (Theorem 2.1): Lonely Runner $\Leftrightarrow |P(e^{i\theta})|$ attains a specific lower bound at some θ .
- Numerical verification to $n = 12$ on $\sim 10^3$ speed sets, including all classical extremal families.

What is not established.

- Lonely Runner for $n \geq 7$ in full generality.
- A new analytic bound on the loneliness gap.

Why Pandrosion does not close it. The Pandrosion reformulation makes Lonely Runner a question about the maximum of a specific trigonometric polynomial. To prove it, one would need either:

1. A real-stability or Pólya–Schur argument on $|P|$, requiring techniques beyond the corpus.
2. A direct construction of θ as a function of $\mathbf{v}$, which the literature has resisted for decades.

References

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